Wind Power: A Practical Overview

Test document 2 of 2 for Vectra

1. Overview

Wind power uses the kinetic energy of moving air to turn the blades of a turbine, which drives a generator. Modern turbines are large: a typical onshore rotor spans 120 to 170 metres, and the biggest offshore machines have rotors longer than two football fields.

In windy regions, onshore wind is the cheapest source of new electricity, with an LCOE of roughly 27 to 55 US dollars per megawatt-hour. Offshore wind is costlier, at roughly 70 to 120 US dollars per megawatt-hour, but delivers more consistent output.

2. How wind plants work

Blades turn a rotor shaft, often through a gearbox, into a generator that produces alternating current. Turbines start generating at wind speeds of about 3 to 4 metres per second and reach full power at about 12 to 14 metres per second. They shut down in storms above roughly 25 metres per second to protect the machine.

Turbines must be spaced several rotor diameters apart so they do not steal wind from each other, which means wind farms cover a large area, though most of that land stays usable for farming.

3. Cost and economics

Onshore wind has an LCOE of roughly 27 to 55 US dollars per megawatt-hour. Offshore wind costs roughly 70 to 120 US dollars per megawatt-hour because of foundations, subsea cables and specialised installation vessels. Turbines typically last 20 to 25 years, and operating costs rise as gearboxes and blades age.

4. Reliability and storage

Wind output varies with weather, but it often complements solar because wind tends to blow more at night and in winter. The capacity factor of onshore wind is typically 35 to 45 percent, and offshore wind reaches 45 to 55 percent, more than double a typical solar plant.

Because wind lulls can last several days, short-duration batteries help less than they do for solar. Grid interconnection, demand flexibility and long-duration storage matter more.

5. Land use and environment

A wind farm's direct footprint is small, about 1 acre per megawatt for roads, pads and foundations, but turbines are spread across 30 to 140 acres per megawatt of surrounding land. Wind produces no emissions while operating, and its lifecycle emissions are roughly 10 to 15 grams of CO2 equivalent per kilowatt-hour.

Concerns include bird and bat collisions, noise and visual impact. Careful siting, radar-triggered shutdowns and blade painting have reduced bird deaths. Blade recycling is an unsolved challenge, because fibreglass blades are hard to separate.

6. Supply chain and jobs

Turbine manufacturing is spread across several regions, including Europe, China and the United States. Wind creates skilled jobs in manufacturing, installation and long-term maintenance, and technicians service turbines for the whole of their lives.

7. Case study: Northgate Offshore Wind Farm (fictional)

The fictional Northgate Offshore Wind Farm has a capacity of 600 megawatts and began operating in 2022. It has 40 turbines, each rated at 15 megawatts, and reaches a capacity factor of 51 percent. It supplies about 650,000 homes.

Outlook: analysts expect onshore wind to keep growing steadily and offshore wind to scale up as ports, vessels and grids are built out.

Note: figures in this document are rounded and illustrative, written as test data.